NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY

9648/01

Paper 1 Multiple Choice

September 2011

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and CT on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

This document consists of **14** printed pages and **0** blank pages.

[Turn over

1 Which pair of organelles has internal membranes?

A Nucleus and lysosome

B Ribosomes and nucleus

C Chloroplast and lysosome

D Chloroplasts and mitochondria

2 An actively growing cell is supplied with radioactive amino acids. Which cell component would first show an increase in radioactivity?

A Nucleus

B Golgi body

C Mitochondrion

D Rough Endoplasmic reticulum

3 Which molecule maintains the fluidity of the cell surface membrane?

A Glycolipid

B Cholesterol

C Glycoprotein

D Phospholipid

4 All of the following are directly associated with the lipid components of the plasma membrane **except**

A Glycerol backbone

B Peripheral proteins

C Hydrocarbon chains

D Cholesterol molecules

5 Which process(s) can take place in a root hair cell when oxygen is **not** available?

A Endocytosis only

B Active transport only

C Diffusion and osmosis

D Active transport and osmosis

6 Which of the following is a feature of exocytosis but not endocytosis?

A Secretion

B Vesicle formation

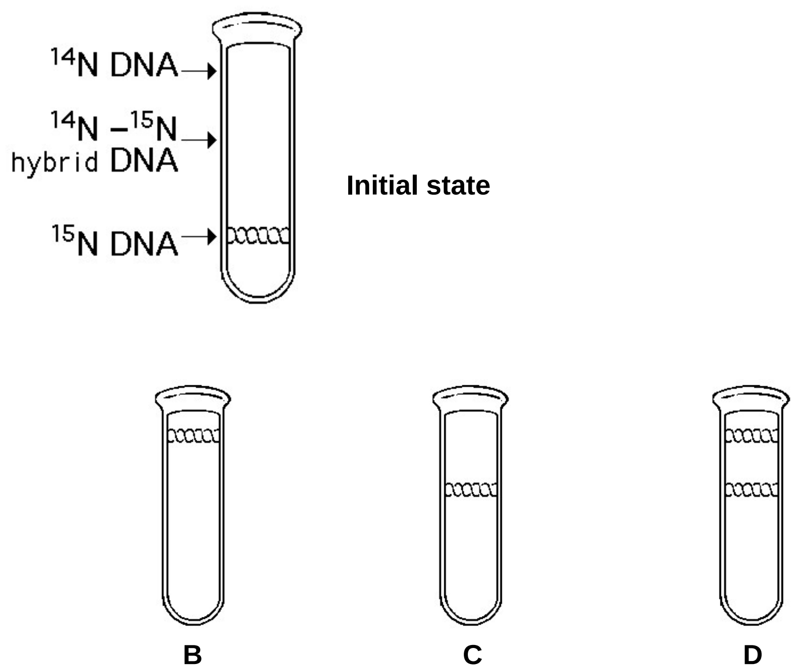
C Lipid bilayer fusion

D Lipid bilayer adhesion

7 Which of the following shows the correct combination of bond(s) that need to be formed and the kind of reaction that is involved in order for the nucleotide to be added to the DNA chain?

	Bond(s) to be formed	Reaction(s) involved
A	Phosphoester	Dephosphorylation
B	Phosphodiester	Hydrolysis
C	Phosphodiester	Condensation and Dephosphorylation
D	Phosphoester and Hydrogen	Condensation

8 In the late 1950s, Meselson and Stahl grew bacteria in a medium containing "heavy" nitrogen (^{15}N) and then transferred them to a medium containing ^{14}N. Which of the following results would be expected after two rounds of DNA replication in the presence of ^{14}N?



9 What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	Difference
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both may not result in disorders.	Gene mutations do not involve inversions but inversion of segments of chromosomes do occur.
C	Both affect DNA sequence.	Chromosomal aberrations may occur across chromosomes but gene mutations occur within a chromosome.
D	Both will not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

10 Which of the following would be the first step in the biosynthesis of a virus with reverse transcriptase?

A Double-stranded RNA must be synthesized.

B A complementary strand of RNA must be synthesized.

C A complementary strand of DNA must be synthesized from a DNA template.

D A complementary strand of DNA must be synthesized from an RNA template.

11 Which of the following statements about bacteria is true?

A Bacteria possess 80s ribosomes, introns and RNA processing.

B Bacteria possess 80s ribosomes, RNA processing and no introns.

C Bacteria possess 70s ribosomes, introns and no RNA processing.

D Bacteria possess 70s ribosomes, no introns and no RNA processing.

12 What is present in a bacteriophage but not in an influenza virus?

A Tail fibres

B Viral capsid

C Glycoprotein

D Reverse transcriptase

13 Which feature occurs in the life cycle of the influenza virus?

A Viruses enter the host cell by endocytosis.

B Host cell DNA is destroyed by lytic enzymes.

C Viral DNA acts as a template for DNA synthesis.

D Viral genome is integrated into the host genome.

14 Four different genes are regulated in different ways.

Gene C: regulatory gene whose product binds to an operator site

Gene D: product undergoes tissue-specific patterns of alternative splicing

Gene E: acetylation and deacetylation occurs to histones binding to the gene

Gene F: part of a group of structural gene controlled by the same regulatory sequence

Which combination correctly identifies which genes are prokaryotic and which are eukaryotic?

	Prokaryotic	Eukaryotic
A	C and F	D and E
B	C and D	E and F
C	D and E	C and F
D	D and F	C and E

15 Which of the following is a feature of eukaryotic gene expression?

A Genes are organized in operons.

B Polycistronic mRNA is common.

C Transcription and translation are spatially separated.

D Translation initiation occurs with a molecule of formyl-methionine.

16 How is translation controlled in eukaryotes?

A By differential removal of introns enabling a gene to code for more than one protein.

B By activation of the protein by folding or cleavage after it is formed.

C By the production of RNA from the non-coding strand of the DNA.

D By protein factors that bind to specific sequences in the mRNA.

17 DNA methylation is known to silence genes because it prevents transcription factors from binding. Which of the following best explains this phenomenon?

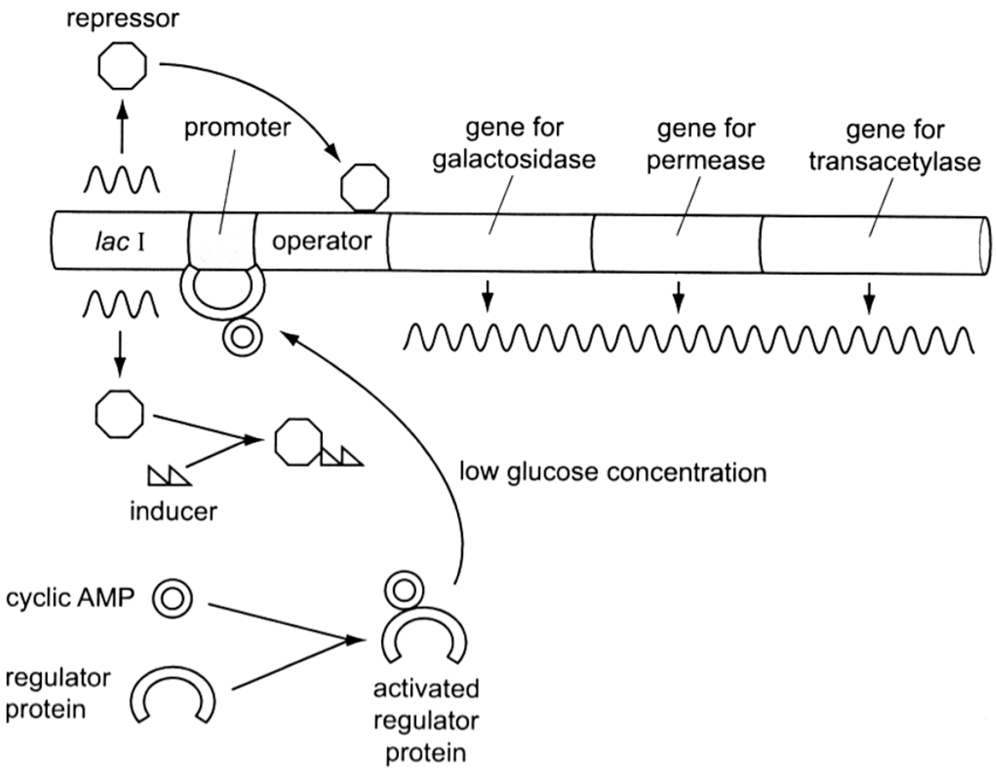
A DNA methylation modifies the shape of the transcription factor.

B DNA methylation prevents dimerization of DNA binding proteins.

C DNA methylation modifies the shape of the DNA element where the transcription factor binds.

D DNA methylation causes acetylation of histone proteins which causes heterochromatin to be formed.

18 The diagram shows some of the factors affecting the lac operon.



Which statement describes the role of the regulator protein in the expression of the lac operon?

- **A** The inactive regulator inhibits transcription.
- **B** The regulator protein is activated when ATP level is high.
- **C** The activated regulator binding stabilizes the transcriptional complex.
- **D** The activated regulator binding to the operator increases mRNA formation.

19 Which pair of features relate to proto-oncogenes and tumour suppressor genes?

	Proto-oncogenes	**Tumour suppressor genes**
A	Trigger cell division	Maintain cell-to-cell adhesion
B	Trigger cell division	Prevent cell cycle arrest
C	Trigger programmed cell death	Trigger cell cycle arrest
D	Trigger programmed cell death	Promote DNA repair mechanism

20 A cross between a tall plant with round leaves and a dwarf plant with round leaves produced the following offspring:

121	round leaves, tall
124	round leaves, dwarf
42	oval leaves, tall
37	oval leaves, dwarf

key	
R	round leaf
r	oval leaf
T	tall
t	dwarf

What were the genotypes of the parents?

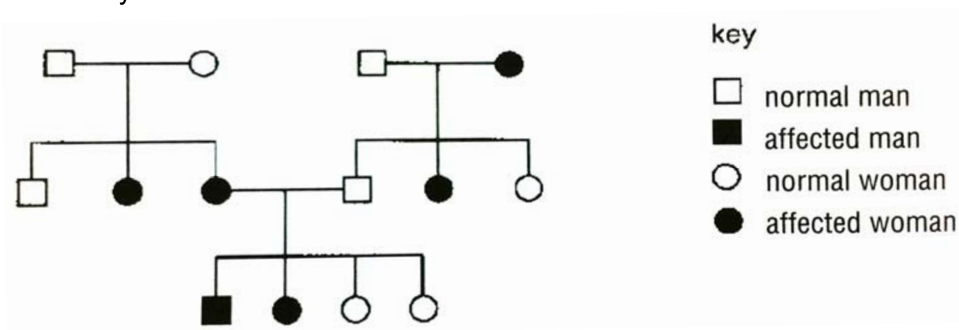
A RrTt x Rrtt

B RrTt x RRtt

C RrTT x Rrtt

D RrTT x RRtt

21 The family tree shows the inheritance of a skin condition.



What is the genetic basis of the skin condition?

A autosomal dominant

B sex-linked dominant

C autosomal recessive

D sex-linked recessive

22 The table summarises a breeding experiment with *Drosophila melanogaster*.

parents	**BBNN x bbnn**			
F1	**BbNn**			
test cross	**BbNn x bbnn**			
test cross offspring	**BbNn**	**bbNn**	**Bbnn**	**bbnn**
number of test cross offspring	596	111	106	465

key

B wild type body colour **N** normal wings

b black body colour **n** vestigial wings

What do these results suggest?

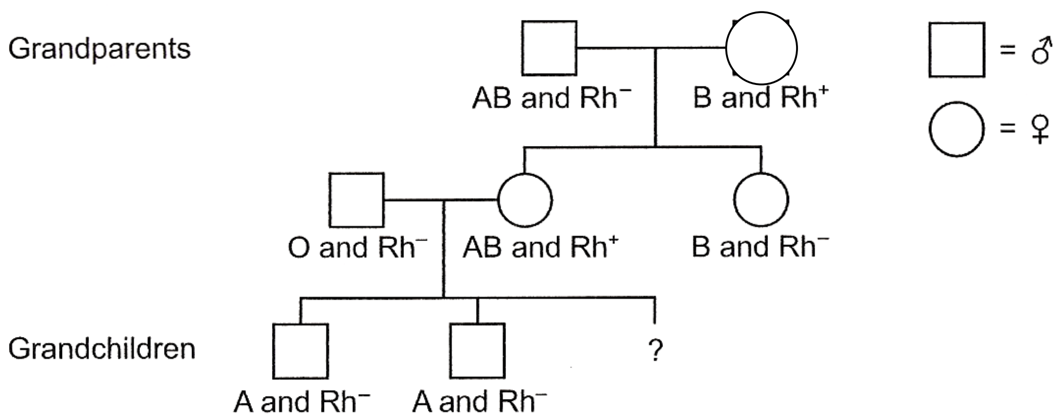
A The alleles **N** and **n** are linked.

B Independent assortment has not occurred.

C The alleles for normal and vestigial wings are codominant.

D The gene loci for body colour and wing shape are not linked.

23 The diagram shows a family tree.



What is the probability that the third grandchild will be a boy with blood group B and Rh positive blood?

A 0.0625

B 0.125

C 0.25

D 0.5

24 In lentils, the seed coat pattern is determined by a gene with 3 alleles. The phenotypes are marbled, spotted and clear. 4 crosses were repeated many times. The crosses and the outcomes of these crosses are shown in the table below.

Cross	Parents	Offspring phenotype and ratio
1	marbled x marbled	3 marbled : 1 clear
2	spotted x clear	1 spotted : 1 clear
3	marbled x marbled	3 marbled : 1 spotted
4	clear x clear	all clear

From the data, it is possible to conclude that

A Spotted is recessive to clear.

B All of the clear offspring are heterozygous.

C Two-thirds of the marbled offspring in cross **3** are heterozygous.

D The marbled parents in cross **1** have the same genotype as the marbled parents in cross **3**.

25 Which one of the following phenotypic features of Man can be affected only by genotype?

A hair colour
B skin colour
C intelligence
D number of different blood group antigens

26 The following reaction sequence occurs in humans.

$$\text{Phenylalanine} \xrightarrow{\mathbf{X}} \text{tyrosine} \xrightarrow{\mathbf{Y}} \text{carbon dioxide and water}$$

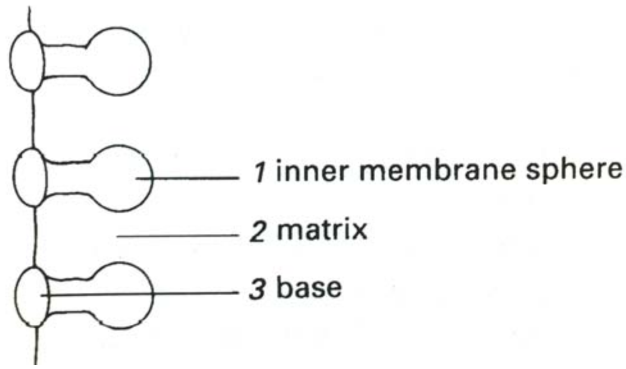
Phenylketonuria (PKU) is a disease caused by an enzyme defiency in step **X**, and alkaptonuria (AKU) is due to an enzyme defiency in step **Y**. Both conditions are rare and caused by recessive alleles.

A person with PKU marries a person with AKU.

What phenotypes would be expected for their children?

A all normal
B all AKU only
C all PKU only
D all both AKU and PKU

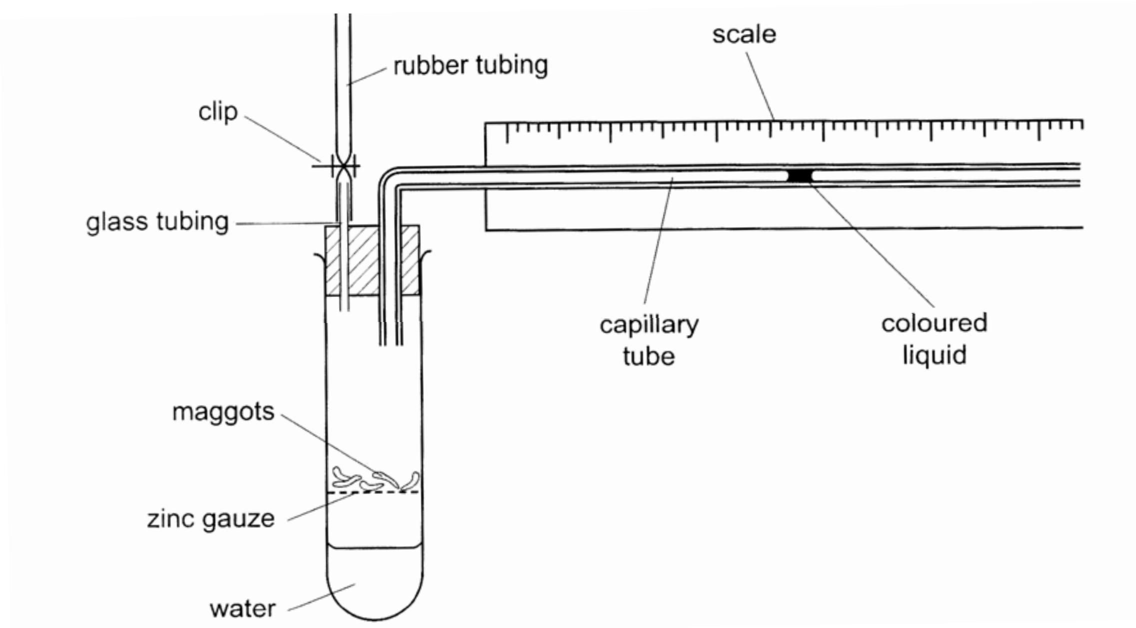
27 The diagram shows stalked particles on part of a crista membrane in a mitochondrion.



What process occurs in each of the numbered regions?

	1	2	3
A	ADP synthesis	electron transport	Krebs cycle
B	ADP synthesis	glycolysis	Krebs cycle
C	ATP synthesis	Krebs cycle	electron transport
D	ATP synthesis	Krebs cycle	glycolysis

28 The diagram shows a respirometer that is being used to investigate respiration in maggots.



The conditions are kept constant and the clip is closed. What happens to the coloured liquid when the maggots respire at an RQ of 1?

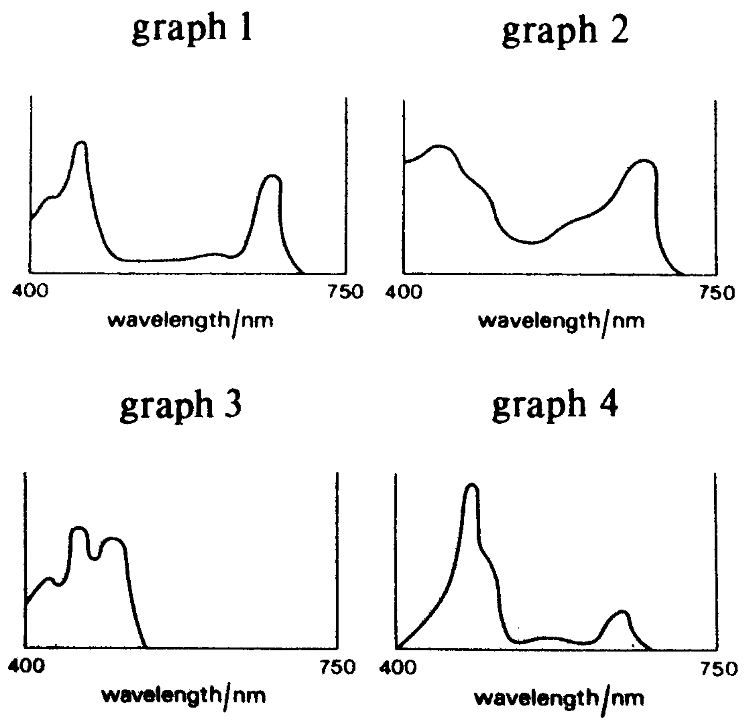
A It moves to the right continuously.

B It moves to the left continuously.

C It moves to the right and stops.

D It remains stationary.

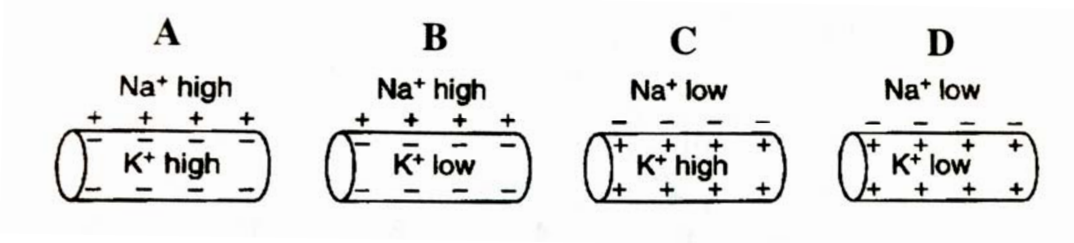
29 Three of the four graphs below show the absorption spectra of photosynthetic pigments. One of the four graphs shows the action spectrum of photosynthesis for a plant containing those pigments.



Which one of the following series identifies the four graphs?

	Absorption spectra			***Action spectrum***
	chlorophyll a	***chlorophyll b***	***carotenoids***	
A	1	4	3	2
B	2	1	3	4
C	2	4	3	1
D	3	2	4	1

30 Which diagram illustrates the distribution of sodium and potassium ions in a section of an axon which is at resting potential?



31 Which of the following statements correctly describes the effect of blood insulin?

	insulin level in blood	effect
A	high	a decrease in glucose metabolism
B	high	an increase in conversion of glycogen to glucose
C	low	decrease in glucose permeability of cells
D	low	an increase in uptake of glucose by muscle cells

32 Signal amplification is most often achieved by

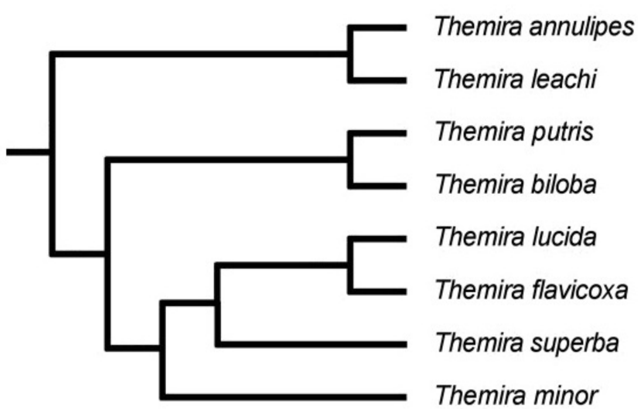
A Branching pathways that produce multiple cellular responses.

B The action of adenylyl cyclase in converting ATP to ADP.

C An enzyme cascade involving multiple protein kinases.

D The binding of multiple signal molecules.

33 The phylogenetic tree below shows the evolutionary relationship between species of the genus *Themira*.



Which of the following statements is **incorrect**?

A All the species are from the Genus *Themira.*

B *Themira minor* is closely related to *Themira superba.*

C *Themira annulipes* and *Themira leachi* descended from a common ancestor.

D *Themira lucida* is more closely related to *Themira flavicoxa* than to *Themira minor.*

34 The mitochondrion is present in all plant and animal cells, but the chloroplast is only present in plant cells. Using phylogeny, what is the best conclusion we can give?

A Mitochondria entered the eukaryotic common ancestor of both plant and animal cells. Chloroplasts entered the plant cell common ancestor after it diverged from the animal cell common ancestor.

B Plant cells can carry out the Calvin's cycle and Krebs cycle with the aid of chloroplasts and mitochondria, whereas animal cells can only carry out the Kreb's cycle with the aid of mitochondria.

C Both mitochondria and chloroplast had entered in the eukaryote common ancestor but the chloroplast left the animal cell common ancestor after the divergence of plant and animal common ancestors.

D The mitochondria and chloroplasts are both examples of eukaryotic cellular organelles that exhibit prokaryote origins; evidence of the endosymbiotic theory.

35 A piece of DNA is introduced into the ampicillin resistant gene of a plasmid. This plasmid also has a tetracycline resistant gene. Bacteria transformed with this plasmid will be

A resistant to both antibiotics

B sensitive to both antibiotics

C resistant to tetracycline only

D sensitive to tetracycline only

36 A student performed the following steps in a Southern blot experiment to determine the number of copies of a particular gene that has been inserted in a genetically modified organism.

i. Transfer of DNA to nitrocellulose membrane.

ii. Restriction digestion of genomic DNA.

iii. Cleaved DNA separated using gel electrophoresis.

iv. Create radioactive probe.

v. Incubate probe and membrane.

Which is the correct sequence?

A ii → iii → i → iv → v

B ii → iii → i → v → iv

C iv → v → i → ii → iii

D v → iv → iii → ii → i

37 Which of the following characteristics is undesirable in cloning vectors used in genetic engineering?

A vulnerable at several sites to a restriction enzyme

B control their own replication

C high copy number

D small in size

38 Which of these processes could increase crop yield to alleviate food shortage problem?

A inserting genes for pest resistance into cotton

B inserting genes for human milk protein into cows

C inserting genes for vitamin A production into rice

D inserting genes for herbicide resistance into corn

39 Which of the following is **NOT** a goal of the Human Genome Project?

A Identify all alleles present in human beings.

B Map the genetic differences between individuals.

C Map and sequence all non-coding sequences in human beings.

D Develop computer software to analyse genetic information which had been sequenced.

40 Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to produce triploid salmon which are infertile.

Reproductive organ tissue from diploid trout was transplanted into the newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish could be seen as genetically modified?

A young trout

B triploid salmon

C salmon providing developing eggs

D trout providing reproductive organ tissue